AI-Powered Research Gap Identifier & Trend Analyzer

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Project Proposal Report

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I. DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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II. ABSTRACT

The rapid growth of scientific publications has created significant challenges for researchers attempting to identify meaningful research gaps during the early stages of research proposal development. Millions of academic articles are published annually across multiple disciplines, making it increasingly difficult for students and early-stage researchers to determine whether their proposed ideas address genuine unresolved problems or unintentionally duplicate existing work. Although academic discovery platforms such as Google Scholar and Semantic Scholar help retrieve related literature, they primarily support retrospective exploration of published research rather than evaluating the novelty of newly proposed research ideas.

Despite advances in Natural Language Processing (NLP), including semantic similarity analysis, topic modeling, and automated gap extraction, current approaches operate largely in isolation and do not provide an integrated framework capable of validating early-stage research ideas against the unresolved limitations of closely related studies.

To address this limitation, this study proposes a Similarity-Guided Semantic Gap Detection Engine that evaluates whether a user-submitted research idea addresses a genuine unresolved gap in existing literature. The system operates within a collaborative research support platform where a similarity retrieval component first identifies relevant papers. The proposed component then extracts key evidence sections (such as methodology, limitations, and future work) and applies transformer-based semantic embeddings and Large Language Model (LLM) reasoning to compare the proposed idea with prior research.

The expected outcome is an explainable system that identifies overlaps, extensions, or genuine research gaps in proposed ideas, helping improve research originality and decision-making. The approach supports SDG 4 (Quality Education) and SDG 9 (Industry, Innovation, and Infrastructure) by promoting higher research quality and reducing redundant development.

Keywords: Research Gap Detection, Semantic Similarity Analysis, Natural Language Processing, Large Language Models, Research Idea Validation.

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My appreciation is also extended to my colleagues and peers for their helpful discussions, suggestions, and feedback during the preparation of this proposal. Their support and collaborative input played an important role in improving the clarity and practicality of the proposed solution.

This research proposal represents an important step toward developing a Similarity-Guided Semantic Gap Detection Engine that supports more accurate, evidence-based research idea validation, contributing to a highly original, rigorous, and efficient proposal development process.

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VII. LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
BERTopic	Bidirectional Encoder Representations Topic Modeling
BiLSTM-CRF	Bidirectional Long Short-Term Memory with Conditional Random Fields
DeBERTa	Decoding-enhanced BERT with Disentangled Attention
DSR	Design Science Research
FR	Functional Requirement
GPU	Graphics Processing Unit
HDBSCAN	Hierarchical Density-Based Spatial Clustering of Applications with Noise
IEEE	Institute of Electrical and Electronics Engineers
IT	Information Technology
LKR	Sri Lankan Rupee
LDA	Latent Dirichlet Allocation
LLM	Large Language Model
ML	Machine Learning
MongoDB	Mongo Database
NLP	Natural Language Processing
NFR	Non-Functional Requirement
PDF	Portable Document Format
REST	Representational State Transfer
RP	Research Project
SaaS	Software as a Service
SBERT	Sentence Bidirectional Encoder Representations from Transformers
SciBERT	Scientific Bidirectional Encoder Representations from Transformers
SDG	Sustainable Development Goal
UI	User Interface
UMAP	Uniform Manifold Approximation and Projection
WBS	Work Breakdown Structure

Table 1: Table of Abbeviations

1. INTRODUCTION

1.1. Background and Context

The rapid growth of scientific publications has created significant challenges for researchers attempting to identify meaningful research gaps within existing literature. Every year, millions of research papers are published across multiple disciplines, making it increasingly difficult for researchers, especially students and early-stage scholars, to determine whether their research ideas address genuine unresolved problems or simply replicate existing work. This problem is particularly critical during research proposal development, where identifying a clear and defensible research gap is essential for producing valuable academic contributions.

This challenge affects multiple groups, including undergraduate students, postgraduate researchers, academic supervisors, and research institutions. Students often struggle to evaluate whether their proposed research ideas are truly novel, while supervisors must manually analyze large volumes of literature to determine whether a proposed topic is sufficiently original and meaningful. This process is time-consuming, cognitively demanding, and increasingly impractical in the context of rapidly expanding scientific output.

Although digital academic platforms now provide access to vast collections of research papers, the availability of literature does not automatically translate into effective gap identification. Academic discovery systems such as Semantic Scholar utilize large-scale literature graphs to organize and retrieve scientific knowledge [2]. However, these systems primarily support retrospective literature discovery and exploration rather than evaluating the novelty of newly proposed research ideas.

Recent advances in Natural Language Processing (NLP) and Large Language Models (LLMs) have significantly improved the automated analysis of scientific literature. Transformer-based models such as SciBERT provide contextualized representations tailored to scientific text and have demonstrated improved performance in scientific NLP tasks [1]. Other research has explored the use of deep contextual language models for measuring semantic similarity across complex technical documents, enabling more accurate comparison of research outcomes and scientific statements [6].

Additional NLP techniques have been developed to analyze large research corpora through topic modeling, semantic clustering, and keyphrase extraction. Topic modeling approaches such as BERTopic combine transformer embeddings with clustering techniques to discover meaningful research themes and trends from scholarly literature [11], [13]. Similarly, phrase embedding approaches such as Key2Vec enable automated extraction of important concepts and keyphrases from research articles [3]. Despite these advances, most current systems still focus on analyzing existing literature rather than evaluating the novelty of early-stage research ideas. Researchers must manually review related papers, identify their limitations, and determine whether a proposed idea meaningfully addresses unresolved challenges.

To address this limitation, this study proposes a Similarity-Guided Semantic Gap Detection Engine capable of evaluating early-stage research ideas through a collaborative workflow. First, a

Similarity/Integrity module retrieves the most relevant academic papers related to a user-submitted idea. Next, the proposed Semantic Gap Detection component performs deeper analysis of the retrieved papers by extracting evidence from their abstract, methodology, limitations, future work, and discussion sections. Using semantic embeddings and LLM-based reasoning, the system then evaluates whether the proposed idea addresses a genuine unresolved gap or merely overlaps with prior work.

By combining similarity-guided literature retrieval, targeted evidence extraction, semantic comparison, and reasoning-based analysis, the proposed system aims to support students and researchers in identifying meaningful research opportunities more efficiently and accurately.

This research aligns with the United Nations Sustainable Development Goals, particularly SDG 4: Quality Education, by improving the rigor and originality of academic research conducted in higher education, and SDG 9: Industry, Innovation and Infrastructure, by supporting innovation through more effective identification of unexplored research opportunities.

The central research question addressed in this study is:

"How can early-stage research gap detection be improved by using similarity-guided semantic analysis to determine if a newly proposed idea addresses a genuine unresolved gap in existing literature?"

1.2. Literature Review

Recent research has explored a range of computational approaches for analyzing scientific literature and identifying meaningful relationships between research documents.One major line of work focuses on the semantic representation of scientific text. Domain-specific transformer models such as SciBERT have demonstrated improved performance in scientific NLP tasks by generating contextual representations tailored to academic literature [1]. These models outperform general-purpose language models in tasks such as document classification, named entity recognition, and semantic similarity detection.

A second major line of work focuses on semantic similarity analysis. Deep contextual language models have been used to compare academic documents and research outcomes, enabling more accurate detection of conceptual similarity beyond simple keyword matching [6]. Similar approaches have also been applied to detect semantic similarity in patent documents using NLP techniques [10]. Other research combines topic modeling with transformer-based embeddings to improve semantic similarity detection in complex textual datasets [8].

Another important area is topic modeling and trend discovery. Traditional methods such as Latent Dirichlet Allocation (LDA) have been widely used to identify major themes in large document collections. Hybrid approaches combining LDA with Word2Vec have been proposed to improve clustering of academic abstracts and enhance semantic topic representation [4]. More recent approaches such as BERTopic leverage contextual embeddings and clustering algorithms such as UMAP and HDBSCAN to identify emerging research themes in large scholarly corpora [11], [13].

Additional research has explored advanced topic modeling techniques using contextualized embeddings for large-scale scientific text analysis [12].

Research has also advanced in keyphrase extraction and evidence identification, where methods such as Key2Vec automatically identify important scientific phrases and concepts from research articles using phrase embeddings [3]. These techniques improve topic interpretation, methodology extraction, and document summarization.

In addition, several studies have investigated automated research gap identification. Machine learning approaches have been used to identify underexplored research topics and emerging trends within scientific literature [5], [14]. Other research has proposed AI-based tools such as GapFinder that analyze limitation and future work sections in research papers to identify unresolved challenges [16]. Earlier studies have also explored text analysis methods for detecting emerging research topics and gaps in scientific fields [9].

Research on novelty detection has further demonstrated that semantic novelty can be modeled through multi-source reasoning, where a new text is compared against multiple prior documents to determine whether it introduces genuinely new information [7].

Despite these advancements, important limitations remain. Most existing approaches focus on analyzing published research documents rather than user-submitted research ideas. Topic modeling systems identify themes and trends but do not determine whether a specific proposed idea fills a research gap. Similarity-based systems retrieve related papers but do not reason over the unresolved limitations of those papers. Gap-detection tools extract potential research gaps from literature but typically do not compare those gaps against newly proposed ideas.

These limitations indicate that existing systems remain largely retrospective, as they analyze completed research rather than supporting predictive validation of early-stage research ideas.

1.3. Research Gap

Although substantial progress has been made in semantic similarity modeling [6], topic modeling [11], [13], keyphrase extraction [3], and automated research gap detection [5], [16], there remains a significant limitation in the ability to evaluate early-stage research ideas against closely related literature**.**

Current approaches primarily focus on one of several isolated tasks:

- Retrieving similar research papers [2]
- Identifying research topics or trends [11], [13]
- Extracting important concepts from academic documents [3]
- Detecting explicit research gaps within published literature [5], [16]

While these techniques are valuable for literature analysis, they do not provide a unified framework capable of determining whether a newly proposed research idea addresses a genuine unresolved gap in the closest existing research.

Existing academic discovery platforms help users find relevant literature, but they do not compare a user-submitted idea with the unresolved limitations and future work described in prior studies. As a result, researchers must still manually synthesize evidence from multiple papers to assess originality and research value.

Current approaches therefore suffer from a retrospective analysis limitation: they analyze existing published content rather than validating early-stage research ideas against the closest relevant evidence.

Therefore, there is a clear need for a system that integrates:

- Similarity-based literature retrieval
- Targeted extraction of methodological and limitation evidence
- Semantic comparison between proposed ideas and prior work
- Reasoning across multiple related research documents

This study addresses that gap by proposing a Similarity-Guided Semantic Gap Detection Engine that combines semantic embeddings, evidence extraction from relevant papers, and LLM-based reasoning to determine whether a proposed research idea fills a genuine unresolved gap or overlaps with existing work.

The novelty of the proposed approach lies in shifting from retrospective corpus-level analysis to idea-level, evidence-based gap validation. By enabling automated assessment of research ideas against the closest existing literature, the proposed system provides a practical tool for supporting early-stage research proposal development.

2. OBJECTIVE

2.1. Main Objective

To design, develop, and evaluate a Similarity-Guided Semantic Analysis & Gap Detection Engine that utilizes Natural Language Processing (NLP) and Large Language Model (LLM) reasoning to accurately assess whether an early-stage, user-submitted research idea addresses a genuine unresolved research gap in closely related academic literature.

2.2. Specific Objectives

To achieve the main objective, the research will be guided by the following measurable specific objectives, which align directly with the component's technical workflow:

1. **Evidence Extraction from Relevant Literature.**
   Extract gap-relevant evidence from the most relevant retrieved research papers, including sections such as the abstract, methodology, limitations, future work, and discussion, to identify information related to existing research challenges and unresolved problems.

2. **Semantic Representation of Research Ideas and Literature**
   Generate semantic representations of both the user-submitted research idea and the extracted literature evidence using transformer-based scientific language models, enabling accurate comparison of conceptual meaning across research texts.

3. **Semantic Alignment and Gap Identification**
   Measure the semantic alignment between the proposed research idea and the unresolved limitations identified in closely related papers in order to detect conceptual overlap, extension of prior work, or potential research gaps.

4. **LLM-Based Reasoning for Gap Assessment**
   Apply Large Language Model (LLM) reasoning to analyze the relationship between the proposed idea and the extracted research evidence, determining whether the idea addresses a genuine research gap, partially extends existing work, or overlaps with prior studies**.**

5. **System Evaluation and Validation**
   Evaluate the effectiveness and practical usefulness of the proposed system using quantitative performance metrics and qualitative expert assessment, measuring the accuracy, relevance, and interpretability of the generated gap-analysis outputs.

3. METHODOLOGY

3.1. Research Approach

This study adopts a Design Science Research (DSR) approach, which focuses on the design and evaluation of technological artifacts that address identified research problems. Design Science is particularly suitable for computing and software engineering research where the objective is to develop and validate a novel system capable of solving a practical problem.

The problem addressed in this research is the absence of systems capable of evaluating whether a newly proposed research idea addresses a genuine unresolved gap in closely related academic literature. Existing tools provide methods for semantic similarity detection [6], [10], topic modeling [11], [12], and automated extraction of research gaps [5], [16] separately, but there is currently no integrated framework that combines similarity retrieval, semantic evidence extraction, and reasoning-based comparison for validating early-stage research ideas.

To address this gap, this research focuses on developing a Similarity-Guided Semantic Gap Detection Engine. The proposed component operates within a larger research support system and works sequentially after the Similarity/Integrity module, which retrieves the most relevant existing research papers related to a user-submitted idea.

The Design Science process in this study follows five stages:

1. **Problem Identification:** Current literature lacks systems capable of validating whether early-stage research ideas address genuine unresolved research gaps [5], [14], [16].

2. **Objective Definition:** Design a semantic analysis framework that determines whether a proposed research idea fills a gap within closely related academic literature.

3. **Artifact Design and Development:** Develop a similarity-guided semantic analysis pipeline that integrates document section extraction, semantic embeddings, and reasoning-based gap evaluation.

4. **Demonstration:** Apply the system to user-submitted research ideas and relevant academic papers retrieved by the similarity module.

Evaluation: Evaluate the system using quantitative metrics and expert-based qualitative assessment.

3.2. Technical Methods and Algorithms

The proposed system follows a similarity-guided semantic analysis pipeline aligned with the supervisor-proposed system workflow. Instead of analyzing an entire literature corpus, the system focuses on a subset of papers retrieved by the similarity module, allowing deeper semantic reasoning over the most relevant prior work. The methodology consists of five main stages:

Stage 1: Similarity-Guided Paper Retrieval

When a user submits a research idea or proposal, the similarity-guided semantic analysis module identifies the most relevant academic papers from the literature database using semantic embedding similarity. Embedding-based semantic retrieval techniques have been widely used to identify conceptually similar research documents beyond simple keyword matching [6], [8], [10].

While the retrieval is executed by the external Similarity module, this handoff acts as the foundational input trigger for the Gap Detection pipeline. By restricting the analysis to this targeted subset, the approach mitigates the computational bottlenecks of large-scale contextual database searches. Large-scale academic retrieval infrastructures such as literature graphs have previously been proposed to support efficient retrieval of related scientific documents [2].

Stage 2: Section-Level Evidence Extraction

Once the similar papers are retrieved, the system extracts sections of each paper that contain information relevant to research gaps and unresolved challenges. The extracted sections include:

- Abstract
- Methodology
- Limitations
- Future Work
- Discussion

These sections are selected because they frequently contain descriptions of research contributions, unresolved limitations, and suggested directions for future work. Prior research on automated research gap detection and literature analysis has demonstrated that limitation and future-work statements often contain explicit descriptions of unresolved research challenges [5], [16].

PDF parsing tools and rule-based heading detection techniques are used to extract structured text segments from the retrieved papers. Sentence segmentation and basic natural language preprocessing are then applied to isolate meaningful evidence statements. The extracted segments form the evidence corpus used for gap analysis.

Stage 3: Semantic Representation

The extracted text segments and the user-submitted research idea are converted into semantic vector representations using transformer-based language models designed for scientific text. Models such as SciBERT, Sentence-BERT (SBERT), DeBERTa-v3 have been specifically trained on large corpora of scientific publications and have demonstrated improved performance on scientific NLP tasks [1]. Other transformer-based models such as contextual embedding systems have also been shown to improve semantic similarity detection in scientific and technical documents [6], [8].

These models generate contextual embeddings that capture semantic meaning within academic literature. By representing both research ideas and extracted paper sections in a shared semantic

vector space, the system can perform accurate comparisons between proposed ideas and existing research.

Stage 4: Asymmetric Semantic Alignment

The system performs asymmetric semantic comparison between the user's proposed research idea and the limitation statements extracted from the retrieved papers in this comparison framework:

- The user's research idea acts as the hypothesis
- The limitations and future work statements from prior papers act as premises

Semantic similarity scores between the idea and extracted evidence statements are calculated using cosine similarity between embedding vectors, a widely used technique for measuring semantic similarity between textual representations [6], [10]. This step determines whether the proposed idea:

- Directly addresses previously identified research limitations.
- Partially extends existing approaches.
- Strongly overlaps with existing research methods.

The asymmetric comparison framework allows the system to evaluate idea-to-gap relationships, rather than simply measuring textual similarity between documents.

Stage 5: LLM-Based Gap Reasoning

In the final stage, a Large Language Model (LLM) performs multi-document reasoning to evaluate the relationship between the user's idea and the extracted research evidence.

Recent studies have demonstrated that LLM embeddings and contextual reasoning can be used to analyze complex textual datasets and support advanced knowledge discovery tasks [15]. LLM-based systems have also been explored for automated research gap identification and scientific literature analysis [14], [16]. The LLM receives the following inputs:

1. User-submitted research idea
2. Extracted methodology summaries from similar papers
3. Extracted limitation and future work statements

Using structured prompts, the model evaluates:

- Unresolved research challenges identified in the literature
- Methodological differences between the proposal and prior research
- Whether the proposed idea addresses an existing research gap.

The LLM generates an interpretable summary describing the relationship between the user's idea and existing literature.

The system categorizes the proposal into one of three outcomes:

- **Gap Addressing** – the idea fills a clearly identified unresolved research gap
- **Partial Overlap** – the idea extends prior work but overlaps with existing methods
- **High Similarity / Low Novelty** – the idea largely duplicates existing research.

This structured output provides an explainable novelty assessment for early-stage research proposals.

3.3. Tools and Platforms with Justification

The system will be implemented using widely adopted tools for Natural Language Processing (NLP) and Machine Learning (ML).

3.3.1. Programming Environment

Python will be used as the primary development language due to its extensive support for machine learning and NLP libraries.

3.3.2. NLP and Embedding Models

Transformer-based models such as SciBERT**,** Sentence-BERT**,** and DeBERTa-v3 will be used to generate semantic embeddings for research text.

Standard language models often struggle with the specialized terminology of scientific documents. SciBERT is specifically designed for scientific text and has been trained on a large corpus of scientific publications, enabling improved representation of domain-specific vocabulary and concepts [1].

Transformer-based contextual embeddings have also demonstrated strong performance in semantic similarity and textual entailment tasks across scientific and technical domains [6], [8].

3.3.3. Large Language Models (LLMs)

LLMs (such as GPT-4o or open-source equivalents like LLaMA-3) will be used via API for multi-document reasoning. LLM-based approaches have recently shown strong potential for analyzing large text corpora, performing semantic reasoning, and supporting advanced topic discovery and literature analysis tasks [14], [15].

3.3.4. Document Processing Tools

PDF parsing and scientific document processing tools will be used to extract structured sections from academic papers. These tools enable reliable extraction of methodological descriptions and limitation statements from research articles.

3.4. Data Collection and Ethical Considerations

3.4.1. Data Collection

The dataset used in this research will consist of open-access academic research papers obtained from publicly available repositories such as:

- arXiv
- Semantic Scholar Open Research Corpus
- CrossRef metadata services

Large-scale academic datasets such as the Semantic Scholar literature graph have been widely used to support research in scientific knowledge discovery and literature analysis [2]. For system evaluation, simulated research proposals will be created to represent early-stage research ideas.

3.4.2. Ethical Considerations

The system analyzes unpublished research ideas, which introduces potential privacy and intellectual property concerns. To mitigate data leakage, user proposals will not be stored permanently, submitted ideas will be anonymized during evaluation, and external LLM APIs will be configured with strict zero-data-retention policies. Furthermore, all outputs will be presented as advisory analysis, not definitive academic judgments.

All research activities will follow institutional ethical guidelines for research involving digital text data.

3.5. Validation Strategy and Performance Metrics

The proposed system will be evaluated using both quantitative and qualitative validation methods**.**

3.5.1. Quantitative Evaluation

The performance of the extraction and similarity components will be evaluated against manually annotated ground-truth datasets using standard information retrieval and statistical metrics:

1. **Precision, Recall, and F1 Score:** These metrics measure the accuracy of correctly extracted gap statements and are widely used in NLP evaluation tasks.

2. **Cosine Similarity Accuracy:** Measures the semantic distance between research ideas and literature using vector representations, a standard approach in semantic similarity research [6], [10].

3. **Correlation measures:** Statistical measures such as Pearson correlation coefficient and Spearman rank correlation will be used to evaluate how strongly the system's automated similarity scores correlate with expert human judgments.

3.5.2. Qualitative Evaluation

Because research gap detection involves complex conceptual reasoning, automated metrics alone are insufficient. Domain experts (such as academic supervisors and researchers) will evaluate system outputs based on:

- Relevance of retrieved literature
- Correctness of identified research gaps
- Clarity of LLM explanations.

Experts will rate outputs using a Likert-scale survey, enabling rigorous assessment of the system's usefulness for research idea validation.

Additionally, expert review will help identify potential LLM hallucinations or reasoning errors, ensuring the reliability of the generated gap analysis.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and its Components

The proposed solution is an AI-driven research support system designed to validate the originality of early-stage research proposals. To manage computational complexity, the platform is architected using a decoupled microservices approach. The architecture consists of four main logical layers:

1. **Client Tier:** Provides the web interface through which users submit research ideas and view analysis results.
2. **Core Application Tier:** Manages system logic, request processing, and communication between components.
3. **AI Processing Component:** The specialized analytical engine deployed in an isolated Python environment. The Semantic Analysis & Gap Detection Engine handles the end-to-end pipeline: retrieving related literature, extracting structural evidence, and evaluating the novelty of the idea.
4. **External Services:** Third-party integrations, including academic literature databases (such as Semantic Scholar API, arXiv) and commercial LLM APIs used for semantic reasoning.

4.2 System Diagram of Components

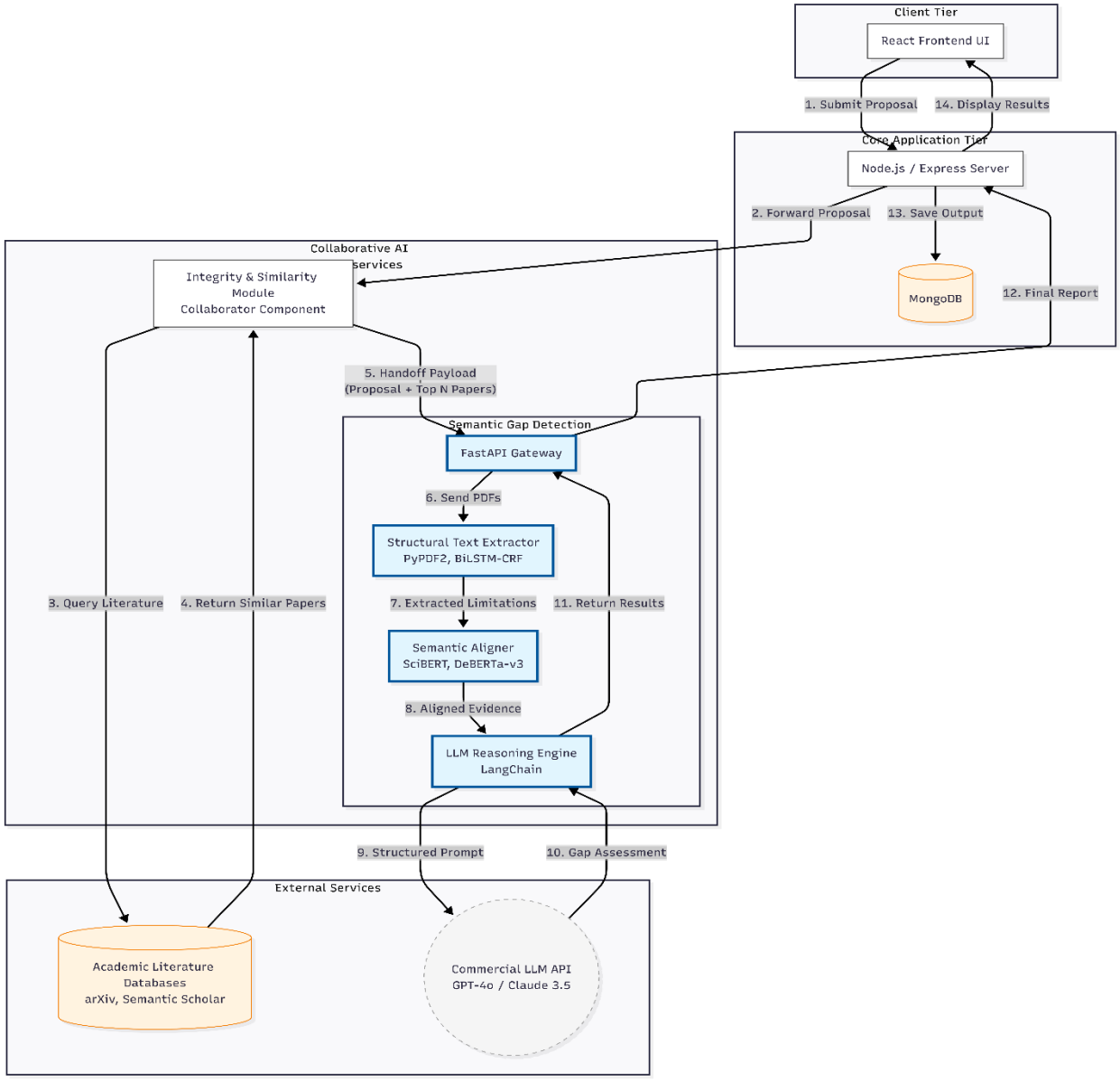


Figure 1: System Diagram (Component Overview)

4.3 Highlight the Individual Component and its Parts

The Semantic Analysis & Gap Detection Engine operates as an isolated, high-performance Python microservice. Its internal architecture is divided into five distinct sub-modules:

1. **Gap Detection API Gateway:** Acts as the entry point for the component and coordinates communication between the application server and the semantic analysis pipeline.
2. **Similarity Retrieval Module:** Encodes the user's raw proposal and queries external academic databases to retrieve a targeted, pre-filtered subset of the top *N* most highly similar published papers.

3. **Structural Text Extractor:** Processes the retrieved academic papers and extracts relevant sections such as methodology, limitations, future work, and discussion that may contain potential research gaps.
4. **Semantic Representation and Alignment Module:** Transforms both the user-submitted research idea and the extracted literature evidence into semantic embeddings using scientific language models. These embeddings allow meaningful comparison between the proposed idea and existing research limitations.
5. **LLM Reasoning Engine:** Uses a Large Language Model to analyze the semantic relationships between the proposed idea and the extracted evidence from multiple papers. The model determines whether the idea fills a genuine research gap, partially overlaps with existing work, or duplicates prior research.

4.4 Component Interactions & Data Flow Explanation

The system's data flow strictly follows a sequential pipeline to optimize accuracy and computational resources:

1. **Ingestion & Routing:** The user submits a raw research proposal via the frontend. The core server routes this text payload directly to the Gap Detection Module.
2. **Similarity Retrieval:** The engine's internal Similarity Module queries external databases and retrieves the most similar academic papers, narrowing the processing scope to only the highly relevant literature.
3. **Extraction & Alignment:** The engine parses these retrieved PDFs, extracting only the critical methodological and limitation-based evidence. It then embeds both the proposal and the evidence to calculate their semantic alignment.
4. **Reasoning Execution:** The highly filtered, relevant text is mapped into a secure prompt and sent to the LLM. The LLM performs multi-document logical reasoning to categorize the novelty of the idea.
5. **Output Generation:** The system returns an interpretable, evidence-backed summary to the core server, which logs the result in MongoDB and displays it to the user.

4.5 Scalability and Security Considerations

- **Scalability:** By decoupling the heavy NLP extraction, embedding, and retrieval algorithms into an independent microservice, the AI backend can be scaled with dedicated GPU cloud instances independently of the web server. Furthermore, the pipeline's design (where the internal similarity module restricts deep extraction exclusively to the top 10-20 papers) eliminates context-window bloat, ensuring rapid processing times and scalable LLM token usage.
- **Security & Data Privacy:** The architecture enforces strict data governance. User proposals are stored transiently in memory during AI processing. Crucially, all API calls made to external LLM providers (such as OpenAI, Anthropic) are configured with strict zero-data-retention headers. This legally and technically guarantees that student research ideas cannot be logged, monitored, or utilized by third parties to train future foundation models.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed Similarity-Guided Semantic Gap Detection System will involve several stakeholders who interact with the system either directly or indirectly.

5.1.1. Primary Stakeholders

1. Students / Early-Stage Researchers
Students and early-stage researchers are the primary users of the system. They submit research ideas or proposal drafts to evaluate whether their idea addresses a genuine research gap or overlaps with existing literature.

2. Academic Supervisors
Supervisors use the system to support the evaluation of student research proposals. The system assists them by summarizing relevant literature evidence and identifying potential research gaps or overlaps.

5.1.2. Secondary Stakeholders

1. Universities and Research Institutions
Universities benefit from improved research quality, better supervision processes, and reduced duplication of research topics.

2. Research Funding Bodies
Funding agencies may benefit from tools that help assess proposal originality and reduce funding for redundant research projects.

3. System Administrators
Administrators maintain the system infrastructure, manage the literature datasets, and ensure system reliability and security.

5.2 Functional Requirements (FR)

FR1: Research Idea Submission
The system shall allow users to submit a research idea, proposal abstract, or project description for analysis.

FR2: Similarity-Based Paper Retrieval
The system shall receive a subset of highly relevant research papers based on the submitted research idea.

FR3: Evidence Section Extraction
The system shall extract relevant sections from the retrieved research papers, including the abstract, methodology, limitations, future work, and discussion.

FR4: Semantic Representation Generation
The system shall convert the user-submitted research idea and extracted literature evidence into semantic representations using transformer-based language models suitable for scientific text.

FR5: Semantic Similarity Analysis
The system shall measure semantic alignment between the proposed research idea and the extracted literature evidence to detect conceptual overlap or potential research gaps.

FR6: Research Gap Reasoning
The system shall apply reasoning-based analysis using Large Language Models (LLMs) to determine whether the proposed idea addresses a genuine research gap, partially overlaps with existing work, or duplicates prior research.

FR7: Gap Assessment Output Generation
The system shall generate an interpretable report summarizing detected gaps, overlaps, and novelty assessments supported by extracted evidence from related research papers.

FR8: Result Presentation
The system shall present analysis results in a clear and understandable format to assist users in evaluating the originality of their research idea.

5.3 Non-Functional Requirements (NFR)

NFR1: Security
The system shall protect user-submitted research ideas and prevent unauthorized access to confidential research proposals.

NFR2: Privacy
The system shall ensure that user-submitted ideas are not permanently stored or shared without authorization, protecting the intellectual property of researchers.

NFR3: Performance
The system shall generate research gap analysis results within an acceptable response time suitable for interactive use.

NFR4: Usability
The system shall provide an intuitive interface that allows users to easily submit research ideas and understand the analysis results.

NFR5: Reliability
The system shall operate consistently and produce analysis results without frequent failures during normal operation.

NFR6: Scalability
The system shall support increasing numbers of users and analysis requests as the platform grows.

NFR7: Maintainability
The system architecture shall allow updates to NLP models, datasets, and reasoning components without major system disruption.

6. COMMERCIALIZATION PLAN

6.1. Target Market and Customer Persona

The proposed Similarity-Guided Semantic Gap Detection System primarily targets organizations and individuals involved in academic research and research proposal development.

6.1.1. Primary Market

The primary market consists of universities, higher education institutions, and research organizations that support undergraduate and postgraduate research programs. These institutions frequently evaluate large numbers of research proposals and can benefit from automated tools that assist in identifying research gaps and validating research originality.

6.1.2. Customer Personas

1. Student Researchers

Undergraduate and postgraduate students developing research proposals are primary end users. They require tools that help them verify whether their research ideas address genuine research gaps before submitting formal proposals.

2. Academic Supervisors

Supervisors are responsible for reviewing multiple student research proposals. The system assists them by providing evidence-based summaries of related research and potential research gaps.

3. Research Committees and Administrators

Institutional research committees and administrators benefit from improved proposal evaluation processes and enhanced research quality within their institutions.

6.2. Value Proposition

The proposed system provides a novel research-support tool that assists users in validating the originality of research ideas during the early stages of proposal development. Key value propositions include:

1. Early research idea validation: Helps students and researchers identify genuine research gaps before investing time in full proposal development.

2. Reduced research duplication: Minimizes unintentional duplication of existing work by analyzing closely related literature.

3. Improved research quality: Supports supervisors and institutions in maintaining higher standards of academic research.

4. Time-efficient literature analysis: Reduces the manual effort required to review multiple research papers when evaluating proposal novelty.

5. Explainable AI-based analysis: Provides evidence-supported summaries that help users understand why an idea may or may not represent a genuine research gap.

6.3. Revenue Model

The system could be commercialized through multiple revenue streams depending on the deployment model.

1. Institutional Licensing Model

Universities and research institutions can subscribe to the platform through annual licensing agreements that provide access for students, supervisors, and research administrators.

2. Software-as-a-Service (SaaS) Model

The system can be deployed as a cloud-based service where users access the platform through a web interface with subscription-based pricing.

3. API Integration Model

The technology could be integrated into existing academic platforms, research management systems, or digital libraries through paid API access.

6.4. Cost Estimation

The primary cost components associated with deploying and maintaining the system include:

1. Infrastructure Costs: Cloud computing infrastructure required for running NLP models, document processing pipelines, and database services.

2. Model and API Usage Costs: Costs associated with large language model inference, semantic embedding models, and third-party API usage.

3. Development and Maintenance Costs: Expenses related to system development, updates to NLP models, software maintenance, and security management.

4. Data Management Costs: Costs associated with maintaining research paper datasets and updating literature sources.

6.5. Pricing Strategy

The pricing strategy may follow a tiered subscription model depending on the number of users and institutional needs.

1. Basic Tier: Access for individual researchers with limited analysis features.

2. Academic Tier: Institution-wide access for universities, supporting multiple students and supervisors.

3. Enterprise Tier: Advanced institutional solutions with API integration and analytics dashboards for research administrators.

This approach allows flexible adoption by different types of users while maintaining scalability.

6.6. Competitive Advantage

The proposed system offers several advantages over existing academic discovery platforms and literature analysis tools.

1. Idea-Level Gap Validation
Unlike existing tools that focus on literature retrieval or trend analysis, the proposed system directly evaluates whether a newly proposed research idea addresses an unresolved research gap**.**

2. Similarity-Guided Semantic Analysis
The system integrates similarity-based literature retrieval with deeper semantic analysis of research limitations, enabling more targeted and meaningful gap detection.

3. Explainable Gap Assessment
The system provides evidence-supported explanations that help users understand the reasoning behind gap identification results.

4. Integration with Research Workflows
The system can operate as part of a larger research support ecosystem, assisting both students and supervisors during proposal development.

6.7. Intellectual Property Considerations

The proposed system involves the development of a novel framework for research idea validation using semantic analysis and gap detection techniques. Possible intellectual property considerations include:

1. Software Copyright

The developed software platform, including its codebase and architecture, can be protected under copyright law.

2. Algorithmic Framework Protection

The unique pipeline combining similarity retrieval, evidence extraction, semantic comparison, and reasoning-based gap validation may form the basis for potential patentable methods depending on jurisdiction.

3. Data Usage Compliance

All academic literature used for system training or evaluation must comply with licensing and open-access policies to avoid copyright infringement.

4. User Data Protection

Since users submit unpublished research ideas, strict privacy protections must be implemented to safeguard intellectual property and confidential research concepts.

7. BUDGET AND JUSTIFICATION

This budget is proposed for developing, training, testing, and deploying the Similarity-Guided Semantic Analysis & Gap Detection component of the research support system. The component involves natural language processing, semantic embeddings, document extraction, and LLM-based reasoning. All costs can be minimized if university laboratory resources and free-tier cloud services are utilized.

7.1. Hardware Costs

Estimated Cost: LKR 0 - LKR 200,000 (optional)

Justification:
If an existing personal laptop or university laboratory computer is available, additional hardware purchases may not be required. However, experimentation with transformer-based models, document processing pipelines, and semantic embeddings may benefit from GPU acceleration.

If necessary, a mid-range laptops or desktops with dedicated GPUs may be used to accelerate experimentation and testing of NLP models. Alternatively, temporary cloud GPU instances can be used for model training and evaluation, which reduces the need for permanent hardware investment.

7.2. Software Licenses

Estimated Cost: LKR 0

Justification:
The proposed system can be developed entirely using open-source software and machine learning frameworks**.** Technologies such as Python, NLP libraries (SpaCy, NLTK), transformer model frameworks (Hugging Face), and open-source databases can be used without licensing fees.

Large Language Model APIs may be used during experimentation, but many providers offer research credits, educational access, or free-tier usage, which can significantly reduce software costs.

If institutional environments require proprietary tools, those licenses are typically covered under existing university agreements.

7.3. Cloud Services

Estimated Cost: LKR 20,000 - LKR 120,000

Justification:
Cloud infrastructure may be required for system testing, deployment demonstrations, and temporary machine learning workloads.

Possible cloud usage includes:

- Hosting the web application, database, and API services for testing and demonstrations.
- Running document processing pipelines and semantic analysis tasks.
- Short-term GPU compute for embedding generation and model experimentation.

Costs can be controlled by using cloud resources only during model training or evaluation periods. Additionally, educational cloud credits from providers such as AWS, Google Cloud, or Azure can further reduce operational costs.

7.4. Human Effort Cost

Estimated Cost: Not billed (student research effort)

Justification:
The design, development, experimentation, documentation, and evaluation of the system will be completed as part of the student's academic research project.

Therefore, development time is not considered a financial cost. If external technical support or specialized consultation becomes necessary, such support is typically provided through the university or academic supervision process.

7.5. Summary Justification

The overall project budget is intentionally kept minimal and cost-efficient**.** The system relies primarily on open-source technologies, publicly available research datasets, and scalable cloud infrastructure**.**

The most significant potential cost arises from cloud computing resources or GPU access required for natural language processing experiments and LMM-based reasoning. However, these costs can be controlled through short-term usage, educational credits, or university computing resources.

As a result, the proposed system can be developed and evaluated within a relatively low-cost research environment while still supporting advanced AI-based experimentation.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1. Task Breakdown Structure

Task ID	Task Name	Description	Timeline
T1	Literature Review	Review research papers on semantic similarity, research gap detection, and AI-based literature analysis.	March 2026
T2	Requirements Analysis	Identify system functional and non-functional requirements.	March 2026
T3	Proposal Preparation	Prepare research proposal and presentation materials.	Before 15 March 2026
T4	System Architecture Design	Design high-level architecture and component structure.	March – April 2026
T5	Dataset Collection	Collect research papers and prepare literature datasets.	April 2026
T6	Similarity Retrieval Module	Develop module to retrieve related research papers.	April – May 2026
T7	Evidence Extraction Module	Implement extraction of methodology, limitations, and future work sections.	May – June 2026
T8	Semantic Gap Detection Engine	Implement semantic alignment and gap reasoning modules.	June – August 2026
T9	System Integration	Integrate system components and prepare for evaluation.	August 2026
T10	System Testing & Evaluation	Evaluate system performance and refine results.	September 2026
T11	Final Report Preparation	Prepare thesis report and project documentation.	September – 11 October 2026
T12	Final System Deployment	Complete website and system submission.	11 October 2026
T13	Final Presentation Preparation	Prepare for final presentation and viva.	12 – 14 October 2026
T14	Research Paper Submission	Prepare and submit research paper.	15 October 2026
T15	Publication Evidence Submission	Submit research paper publication evidence.	1 December 2026

Table 2: Work Breakdown Structure

8.2. Timeline Alignment

Phase	Period	Activities
Phase 1	March 2026	Literature review, requirements analysis, proposal preparation and submission
Phase 2	March – May 2026	System architecture design and initial module development (Progress Presentation 1)
Phase 3	May – August 2026	Development of semantic gap detection engine and system components (Progress Presentation 2)
Phase 4	August – September 2026	System integration, testing, and evaluation
Phase 5	September – October 2026	Final report preparation and system deployment
Phase 6	October – December 2026	Final presentation, research paper submission, and publication evidence submission

Table 3: Timeline Alignment

9. GANTT CHART

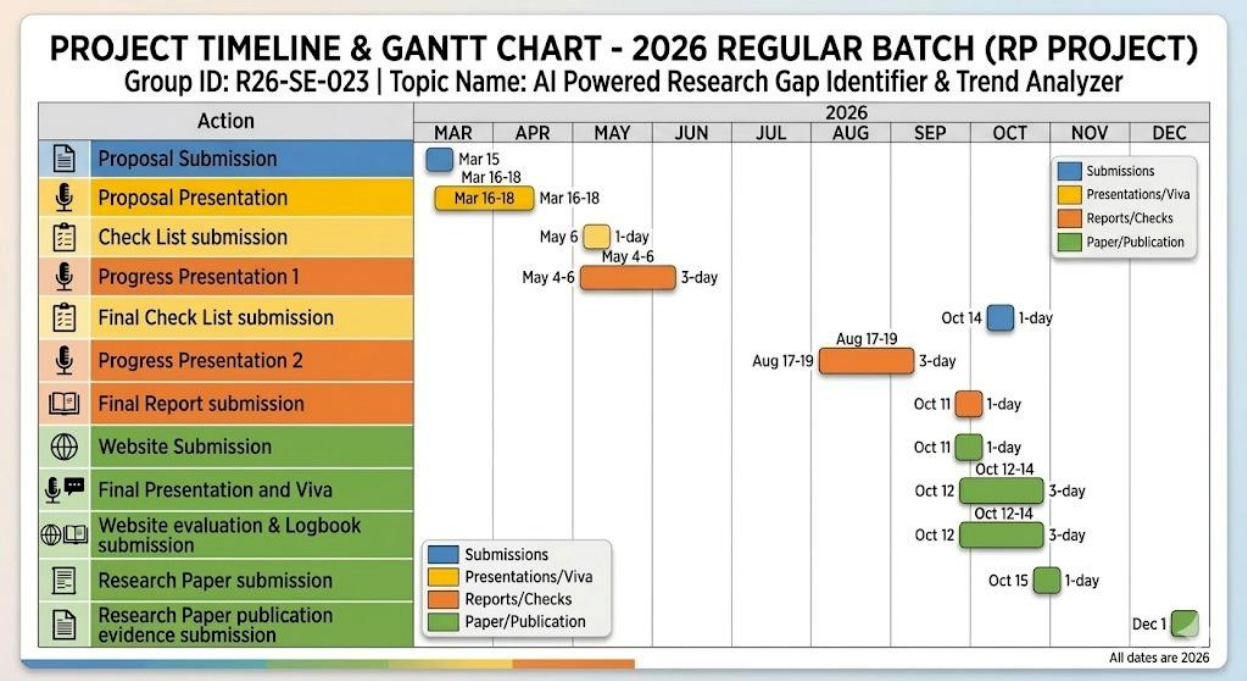


Figure 2: Gantt Chart

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11. APPENDICES

11.1 Detailed diagrams

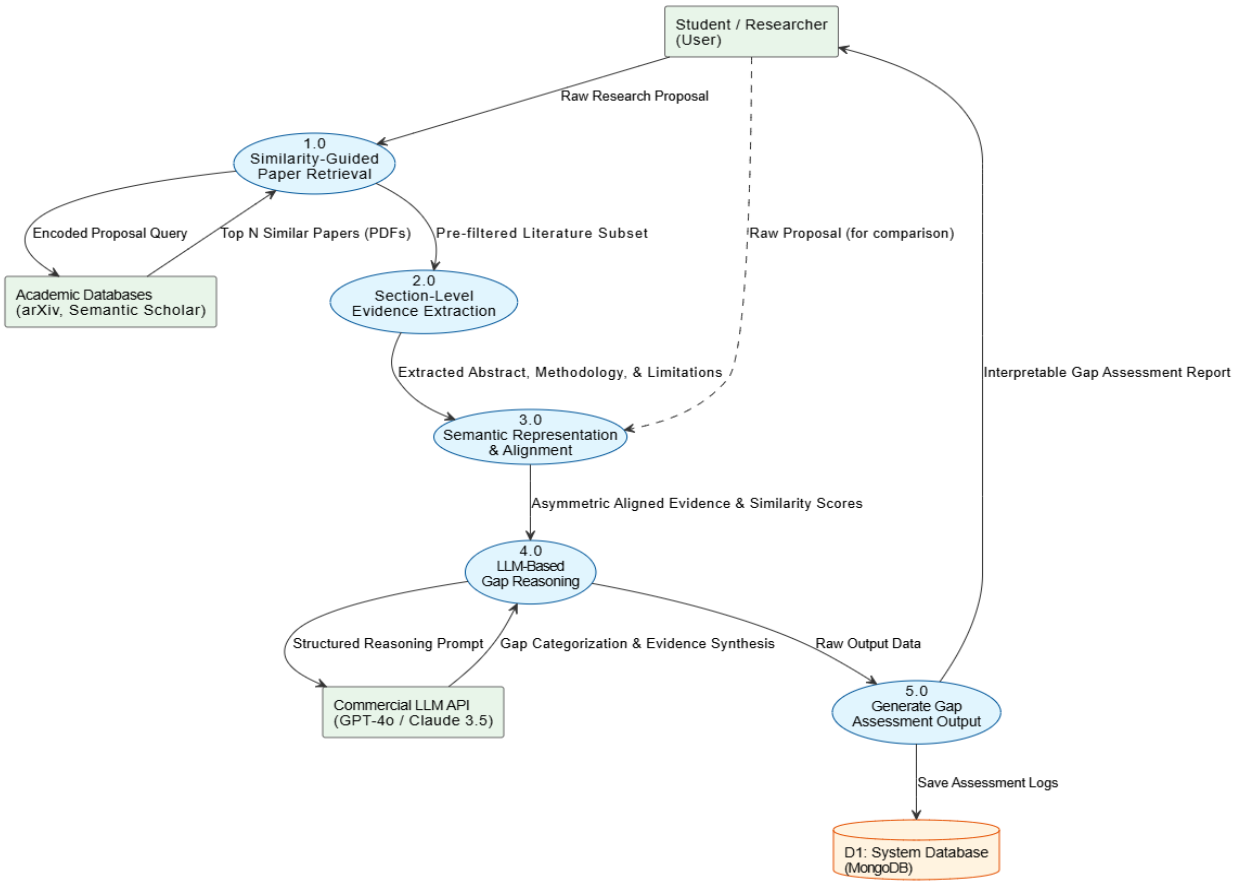


Figure 3: Data Flow Diagram

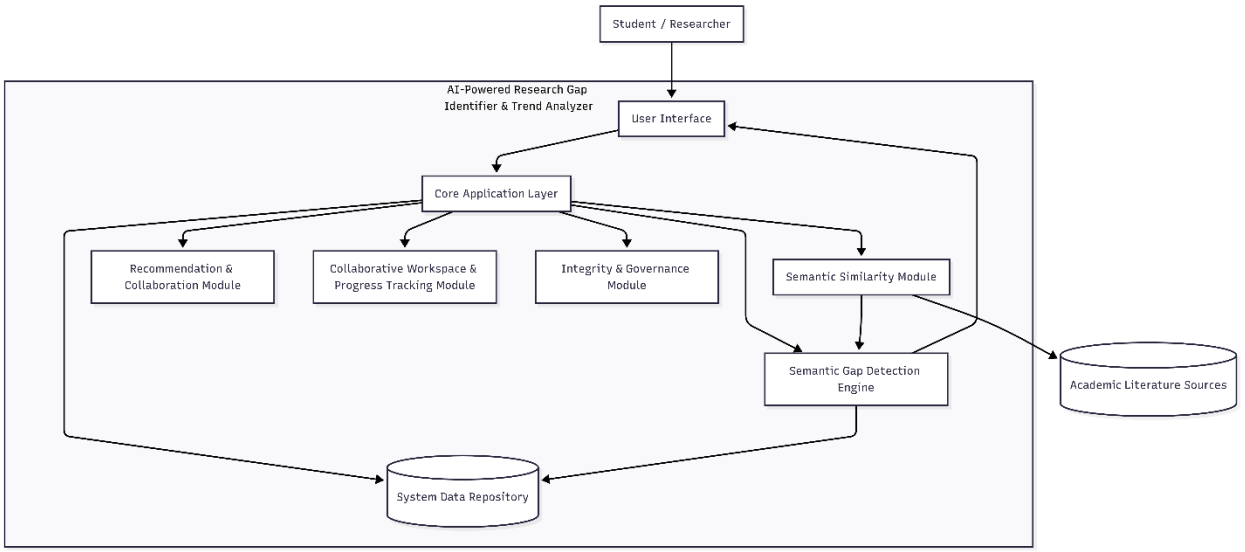


Figure 4: System Diagram (Overall)

11.2. Risk Analysis

Risk	Mitigation
LLM hallucination	Use structured prompts that are grounded by extracted evidence
PDF extraction errors	Rule-based section detection
Lack of labeled data for evaluation	Use expert validation + synthetic dataset
High computational cost (LLM + embeddings)	Optimize with top-N retrieval + batching
Bias in training data	Use diverse datasets + evaluation checks

Table 4: Risk Analysis Table